

18/08/25

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Differential Equation

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Simultaneous differential equation

$$\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}$$

2023

MS

solve: $\frac{dx}{dt} = ny - mz$ $\frac{dy}{dt} = lz - nx$

$$\frac{dz}{dt} = mx - ly$$

$$\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r} = dt$$

$$\frac{dx}{ny - mz} = \frac{dy}{lz - nx} = \frac{dz}{mx - ly} = dt$$

using multiplier method

denominator zero.

 x, y, z - multiplier

$$\frac{x dx}{nxy - mxz} = \frac{y dy}{lyz - nxy} = \frac{z dz}{mxz - lyz}$$

$$\frac{x dx + y dy + z dz}{nxy - mxz + lyz - nxy + mxz - lyz}$$

$$x dx + y dy + z dz = 0$$

Integrating

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C_1$$

$$\boxed{x^2 + y^2 + z^2 = C_1} \rightarrow \text{equ (1)}$$

l, m, n mult. ...

$$\frac{l dx}{l y - l m z} - \frac{m dy}{l m z - m n x} = \frac{n dz}{m n x - l n y}$$

$$\frac{l dx + m dy + n dz}{l y - l m z + l m z - m n x + m n x - l n y}$$

$$l dx + m dy + n dz = 0$$

$$\boxed{l dx + m dy + n dz = C_2}$$

Q18

②

Solve $\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{z}$

$$x dx = y dy \quad \frac{dy}{dx} = \frac{dz}{z}$$

$$x dx - y dy = 0$$

$$\frac{x^2}{2} - \frac{y^2}{2} = C_1$$

$$\boxed{x^2 - y^2 = C_1}$$

$$x = \sqrt{y^2 + C_1}$$

$$\frac{dy}{\sqrt{y^2 + C_1}} = \frac{dz}{z}$$

$$\log \{y + \sqrt{y^2 + C_1}\} = \log z + \log C_2$$

$$\log A + \log B = \log AB$$

$$\log \{y + \sqrt{y^2 + C_1}\} = \log z C_2$$

$$\boxed{y + \sqrt{y^2 + C_1} = z C_2}$$

Que 3: Method of multiplier

$$\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} = \frac{dz}{x^2+y^2}$$

$$\frac{dx}{xz+yz} = \frac{dy}{xz-yz} = \frac{dz}{x^2+y^2}$$

y, x, z multiplier

$$\frac{y dx}{xyz + y^2 z} = \frac{x dy}{x^2 z - xy z} = \frac{-z dz}{-x^2 z - y^2 z}$$

$$\frac{y dx + x dy - z dz}{xyz + y^2 z + x^2 z - xy z - x^2 z - y^2 z}$$

$$y dx + x dy - z dz = 0$$

$$xy - \frac{z^2}{2} = C_1$$

$$\boxed{2xy - z^2 = 2C_1}$$

$$x, -y, -z \quad \frac{x dx}{x^2 z + xy z} = \frac{-y dy}{-xy z + y^2 z} = \frac{-z dz}{-x^2 z - y^2 z}$$

$$\frac{x dx - y dy - z dz}{x^2 z + xy z - xy z + y^2 z - x^2 z - y^2 z}$$

$$x dx - y dy - z dz = 0$$

$$\frac{x^2}{2} - \frac{y^2}{2} - \frac{z^2}{2} = C_2$$

$$\boxed{x^2 - y^2 - z^2 = 2C_2}$$

$$a_1x + a_2y = b_1$$

$$a_3x + a_4y = b_2$$

$$x = \dots, y = \dots$$

Que ⑤

$$Dx - 7x + y = 0 \quad \text{--- ①}$$

$$Dy - 2x - 5y = 0 \quad \text{--- ②}$$

23/8 ⑤
Que 2022

$$\frac{dx}{dt} - 7x + y = 0$$

$$\frac{dy}{dt} - 2x - 5y = 0$$

$$(\mathbb{D} - 7)x + y = 0$$

$$-2x + (\mathbb{D} - 5)y = 0 \quad \text{--- ②}$$

$$(\mathbb{D} - 5)(\mathbb{D} - 7)x + (\mathbb{D} - 5)y = 0$$

$$-2x + (\mathbb{D} - 5)y = 0$$

$$(\mathbb{D} - 7)(\mathbb{D} - 5)x + 2x = 0$$

$$(\mathbb{D}^2 - 5\mathbb{D} - 7\mathbb{D} + 35 + 2)x = 0$$

$$(\mathbb{D}^2 - 12\mathbb{D} + 37)x = 0$$

$x = CF$

$$m^2 - 12m + 37$$

$$m^2 - 6m - 2m + 37$$

$$m = \frac{12 \pm \sqrt{144 - 4(37)}}{2}$$

$$m = \frac{12 \pm \sqrt{144 - 148}}{2}$$

$$m = \frac{12 \pm \sqrt{-4}}{2}$$

$$m = \frac{12 \pm i2}{2}$$

$$m = 6 \pm i$$

Right
 $x = CF + PT$

$m = \text{Complex}$

$$m = \alpha + i$$

$$m = \alpha + i, \alpha - i$$

$$x_m = \boxed{e^{\alpha t} (C_1 \cos t + C_2 \sin t)}$$

$$\Rightarrow \frac{dx}{dt} - tx + y = 0 \quad \text{--- (1)}$$

$$\begin{aligned} &= \frac{d}{dt} e^{\alpha t} (C_1 \cos t + C_2 \sin t) - t e^{\alpha t} (C_1 \cos t + C_2 \sin t) + y \\ &= e^{\alpha t} (C_1 \cos t + C_2 \sin t) + e^{\alpha t} [-C_1 \sin t + C_2 \cos t] \\ &\quad - t e^{\alpha t} (C_1 \cos t + C_2 \sin t) + y = 0 \end{aligned}$$

$$- e^{\alpha t} (C_1 \cos t + C_2 \sin t) + e^{\alpha t} [-C_1 \sin t + C_2 \cos t] + y = 0$$

$$e^{\alpha t} (-C_1 \cos t + C_2 \sin t - C_1 \sin t + C_2 \cos t) + y = 0$$

$$e^{\alpha t} \{ (-C_1 + C_2) \cos t + (-C_2 - C_1) \sin t \} + y = 0$$

$$e^{\alpha t} \{ C_3 \cos t + C_4 \sin t \} + y = 0$$

$$y = e^{\alpha t} \{ -C_3 \cos t - C_4 \sin t \}$$

$$y = e^{\alpha t} \{ C_3 \cos t + C_4 \sin t \} \quad \underline{\underline{\text{Ans}}}$$

$$t \frac{dx}{dt} = t - 2x$$

$$t \frac{dy}{dt} = tx + ty + 2x - t$$

Civil
2020
Que 6

equation ① $\frac{dx}{dy} = 1 - \frac{2x}{t}$

$\frac{dx}{dy} + \frac{2}{t}x = 1$

linear diff. eqn

I.F.

$e^{\int \frac{2}{t} dt} = e^{+2 \log t}$
 $= e^{\log t^2}$

I.f:

$= t^2$

$t^2 x = \int t^2 dt + C$

$t^2 x = \frac{t^3}{3} + C$

$x = \frac{t}{3} + \frac{C_1}{t^2}$

equation ②

$t \frac{dy}{dt} = tx + ty + 2x - t$

$\frac{dy}{dt} = x + y - \frac{1}{t}$

by equation ①

$\frac{dy}{dt} = x + y - \frac{dx}{dt}$

$\frac{dy}{dt} + \frac{dx}{dt} = x + y$

$\frac{dx}{dt} + \frac{dy}{dt} = x + y$

$$\frac{dx + dy}{x+y} = dt$$

$$\log(x+y) = t + \log C$$

$$\log \frac{x+y}{C_2} = t$$

$$\frac{x+y}{C_2} = e^t$$

$$x+y = C_2 e^t$$

$$y = C_2 e^t - x$$

$$y = C_2 e^t - \frac{t}{3} - \frac{C_1}{t^2} \quad \underline{\underline{Ans}}$$

2023 Que 7

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a\sqrt{x^2+y^2+z^2}}$$

$$\frac{dx}{x} = \frac{dy}{y}$$

$$\log x = \log y + \log C_1$$

$$\frac{x}{y} = C_1$$

x, y, z multiplier.

$$\frac{x dx + y dy + z dz}{x^2 + y^2 + z^2 - a z (\sqrt{x^2+y^2+z^2})}$$

$$x^2 + y^2 + z^2 = t^2$$

$$x dx + y dy + z dz = t dt$$

$$= \frac{t dt}{t^2 - a^2 z^2}$$

$$= \frac{dt}{t - a^2 z}$$

$$\Rightarrow \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a\sqrt{x^2 + y^2 + z^2}} = \frac{dt}{t - a^2 z}$$

$$\textcircled{1} \quad \frac{dx}{x} = \textcircled{2} \quad \frac{dy}{y} = \textcircled{3} \quad \frac{dz}{z - a^2 t} = \textcircled{4} \quad \frac{dt}{t - a^2 z}$$

$\textcircled{3} + \textcircled{4}$

add $\textcircled{3} + \textcircled{4}$

$$\frac{dz + dt}{z + at + t - a^2 z} \Rightarrow \frac{dz + dt}{(z + t) - a(z + t)} \Rightarrow \frac{dz + dt}{(z + t)(1 - a)}$$

$$\frac{dx}{dx} + \frac{dy}{y} = \frac{dz}{z - a\sqrt{x^2 + y^2 + z^2}} = \frac{dt}{t - a^2 z} = \frac{dz + dt}{(z + t)(1 - a)}$$

$\textcircled{1} \quad \textcircled{2} \quad \textcircled{3} \quad \textcircled{4} \quad \textcircled{5}$

$\textcircled{1}$ and $\textcircled{5}$

$$(1 - a) \frac{dx}{x} = \frac{dz + dt}{z + t}$$

$$(1 - a) \log x = \log(z + t) + \log C_2 + x^{1-a}$$

$$x^{1-a} = (z + t) C_2$$

$$C_2 = \frac{x^{1-a}}{z + t}$$

2020
Civil

Lagrange's form

$$Pp + Qq = R$$

$$\left[\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \right] \Rightarrow \text{Simultaneous eqn diff.}$$

$$f(c_1, c_2) = 0$$

Most
imp. Ques

Lagrange's form
simultaneous

$$p + 3q = 5z + \tan y - 3x \quad \text{--- la.}$$

simultaneous

$$\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)}$$

I and II

$$\frac{dx}{1} = \frac{dy}{3}$$

$$3dx = dy$$

$$0 = dy - 3dx$$

$$C_1 = y - 3x$$

$$\left\{ \begin{array}{l} \therefore Pp + Qq = R \Rightarrow \\ \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \\ f(c_1, c_2) = 0 \end{array} \right.$$

I and III

$$\frac{dx}{1} = \frac{dz}{5z + \tan C_1}$$

$$x = \frac{\log(5z + \tan C_1)}{5} + \frac{C_2}{5}$$

$$5x = \log(5z + \tan C_1) C_2$$

$$e^{5x} = (5z + \tan C_1) C_2$$

$$\boxed{\frac{e^{5x}}{5z + \tan(y-3x)} = C_2}$$

$$f\left(y-3x, \frac{e^{5x}}{(5z + \tan(y-3x))}\right) = 0$$

Total differential equation

$$P dx + Q dy + R dz = 0$$

method ① Inspection method

② Taking one variable constant

30/8/21 chem ③ homogeneous

2014 ④ Method of Auxiliary equation

Que ① $(2x^2 + 2xy + 2xz^2 + 1)dx + dy + z dz = 0$

Que 2018 ② $(y^2 + yz)dx + (xz + y^2)dy + (y^2 - xy)dz = 0$

Que 2019 ③ $3x^2 dx + 3y^2 dy - (x^3 + y^3 + e^{2z}) dz = 0$

Condition of Integrability

$$P dx + Q dy + R dz = 0$$

$$\begin{array}{ccc} P & Q & R \\ x & y & z \end{array}$$

$$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0$$

Que ④ solve $(y+z)dx + dy + dz = 0$ — ①

condition of integrability $Pdx + Qdy + Rdz = 0$

$$(y+z) \begin{matrix} y+z \\ x \end{matrix} [0-0] + 1 \begin{matrix} 1 \\ y \end{matrix} [0-1] + 1 \begin{matrix} 1 \\ z \end{matrix} [1-0] \\ -1+1=0$$

condition of integrability is satisfied.
- Taking one variable constant
 $x = \text{constant}$

$$dx = 0$$

$$(y+z) \underset{\downarrow 0}{dx} + dy + dz = 0$$

$$dy + dz = 0$$

Integrating $y+z = f(x)$ — ②
differentiating

$$dy + dz = f'(x) \text{ — ③}$$

from equation ① $dy + dz = (-y+z)dx$

Comparing eqn ① and ②

$$(-y+z)dx = f'(x) \text{ — ④}$$

equation ② $y+z = f(x)$

$$-f(x) = f'(x)$$

$$f'(x) + f(x) = 0$$

linear
diff. equ.

$$\frac{d}{dx} f(x) + f(x) = 0$$

$$(D+1)f(x) = 0$$

$$m+1=0$$

$$m = -1$$

C.f.

$$e^{-1x}$$

$$\boxed{f(x) = Ce^{-x}}$$

equ ② से

$$y+z = Ce^{-x}$$

Ques ⑤
imp Ques ⑤ $3x^2 dx + 3y^2 dy - (x^3 + y^3 + e^{2z}) dz = 0$

$C \cos x + e^{xy} dx + (e^x + e^y) dy + e^z dz = 0$

Ques ⑥ $(e^{xy} + e^z) dx + (e^y dz + e^x) dy + (e^y - e^{xy} - e^z) dz = 0$

civil 2023 Ques ⑥ $p + 3q = 5z + \tan(y - 3x)$

2023 ⑨ $\frac{dx}{x} + \frac{dy}{y} = \frac{dz}{z - a\sqrt{x^2 + y^2} + z^2}$

2020 ⑩ $\frac{dx}{dt} = t - 2x$
 $\frac{dy}{dt} = tx + ty + zx = t$

11/8/2025 Method of forming Auxiliary Eqn

2019 ① $3x^2 dx + 3y^2 dy \rightarrow x^3 + y^3 + e^{2z} + dz = 0$

P	Q	R
x	y	z

$\frac{dx}{\frac{\partial P}{\partial z} - \frac{\partial R}{\partial y}} = \frac{dy}{\frac{\partial R}{\partial y} - \frac{\partial P}{\partial z}} = \frac{dz}{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}$

$\frac{dx}{0 + 3y^2} = \frac{dy}{-3x^2 - 0} = \frac{dz}{0 - 0}$ ②

$\frac{dx}{3y^2} = \frac{dy}{-3x^2} = \frac{dz}{0}$ ②

$$-3x^2 dx = 3y^2 dy$$

$$-x^2 dx = y^2 dy + C \rightarrow -\frac{x^3}{3} + \frac{y^3}{3} = C_1$$

$$x^3 + y^3 = C_1 \pm u - C$$

$$Adu + Bdv = 0$$

$$u \rightarrow du$$

$$3x^2 dx + 3y^2 dy = du \rightarrow (4)$$

III

$$dz = 0$$

$$z = C_2' = v \rightarrow (5)$$

$$dz = dv \rightarrow (6)$$

$$A(3x^2 dx + 3y^2 dy) \rightarrow (7)$$

$$3x^2 A dx + 3y^2 A dy + B dz = 0$$

$$3x^2 dx + 3y^2 dy + -(x^3 + y^3 + e^{2z}) dz = 0 \rightarrow (8)$$

Comparing ① & ② $\rightarrow A = 1$

$$Adu + Bdv = 0$$

$$du - (x^3 + y^3 + e^{2z}) dv = 0$$

$$du \rightarrow (u + e^{2v}) v = 0$$

②

$$(y+3)dx + dy + dz = 0$$

$$\frac{dx}{0} = \frac{dy}{-1} = \frac{dz}{1} \rightarrow (9)$$

$$dy = -1$$

$$dz = 1$$

$$dy = -dz$$

$$y = -z + C_2$$

$$y + z = C_2 = v$$

$$dx = 0$$

$$x = C_1 = u$$

$$du = dx$$

$$Adu + Bdv = 0$$

$$A(dv) + B(dy + dz) = 0$$

$$\boxed{A=1} \quad \boxed{B=y+z}$$

$$Vdu + dv = 0$$

$$v + u' = -dv$$

$$du = \frac{-dv}{V}$$

$$\boxed{u = -\log V + \log c}$$

$$u = \log \frac{c}{V}$$

$$e^u = \frac{c}{V}$$

$$V = \frac{c}{e^u} \quad \boxed{y+z = ce^{-x}}$$

Que $(2x^2 + 2xy + 2xz^2 + 1)dx + dy + z dz = 0$
6/9/2025 inspection method.

Que $(yz + 2x)dx + (xz - 2z)dy + (xy - 2y)dz = 0$

Condition of integrability is satisfied

$\begin{matrix} P & Q & R \\ x & y & z \end{matrix}$

$$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0$$

$$\textcircled{1} \text{ } d(xy)$$

$$① d(xyz) = yz dx + xz dy + xy dz$$

$$② dx^2 = 2x dx$$

$$③ d(yz) = z dy + y dz$$

* Method of Homogeneous.

Que ① $(y^2 + yz)dx + (xz + z^2)dy + (y^2 - xy)dz = 0$ — ①

$$Px + Qy + Rz = 0$$

$$(y^2 + yz)x + (xz + z^2)y + (y^2 - xy)z = 0$$

$$xy^2 + xyz + xyz + yz^2 + y^2z - xyz \neq 0$$

$$xy^2 + xyz + yz^2 + zy^2 \neq 0$$

$$xy(y+z) + yz(z+y) \neq 0$$

$$(y+z)(xy+yz) \neq 0$$

$$I = y(y+z)(x+z) \neq 0 \text{ — ②}$$

$$\frac{1}{Px + Qy + Rz}$$

$$\frac{\text{Equation 1 (Given Equation)}}{I (\text{Equation 2})}$$

$$dI = \text{Given Equation} + \text{Something}$$

$$dI - \text{Something} = \text{Given Equation}$$

$$dI = dy(x+z)(y+z) + y(dx+dz)(y+z) + y(x+z)(dy+dz)$$

Given equation

$$(y^2+yz)dx + (xz+z^2)dy + (y^2-xy)dz$$

$$\Rightarrow y^2 dx + yz dx + xz dy + z^2 dy + y^2 dz - xy dz$$

$$dI = xy dy + xz dy + yz dy + z^2 dy + (y^2+yz)(dx+dz) + (xy+yz)(dy+dz)$$

$$dI = xy dy + xz dy + yz dy + z^2 dy + y^2 dx + y^2 dz + yz dx + yz dz + xy dy + xy dz + yz dy + yz dz - xy dz + xy dz$$

$$dI \Rightarrow y^2 dx + yz dx + xz dy + z^2 dy + y^2 dz - xy dz + [2xy dy + 2yz dy + 2yz dz + 2xy dz]$$

$$dI - [7 \text{ } 8 \text{ } 9 \text{ } 10] =$$

$$dI - 2xy dy - 2yz dy - 2yz dz - 2xy dz = \text{Given equ.}$$

$$dI - 2y(x+z)dy - 2y(x+z)dz = \text{Given equ.}$$

— (9)

Equation = $\frac{dI - 2y(x+z)dy - 2y(x+z)dz}{I}$

$$\frac{dI}{I} - \frac{2y(x+z)dy - 2y(x+z)dz}{I}$$

$$\frac{dI}{I} - \frac{2y(x+z)(dy+dz)}{y(x+z)(y+z)} = 0$$

$$\log I - 2 \log(y+z) = \log c$$

$$\log \frac{I}{(y+z)^2} = \log c$$

$$I = c(y+z)^2$$

$$Px + Qy + Rz = c(y+z)^2$$

$$y(x+z)(y+z) = c(y+z)^2$$

$$\boxed{y(x+z) = c(y+z)}$$

Home
★

Que (2) $(yz + z^2)dx - xz dy + xy dz = 0$

$$Px + Qy + Rz \neq 0$$

$$(yz + z^2)x + xzy + xyz \neq 0$$

$$xyz + xz^2 + xzy + xyz \neq 0$$

$$2xyz + xz^2 \neq 0$$

$$xz(2y+z) \neq 0$$

$$\frac{1}{Px + Qy + Rz} \quad \text{Given equation}$$

$$\frac{dI}{I} - \frac{2xz dy - 2xz dz}{I} = 0$$

$$\frac{dI}{I} - \frac{2xz dy - 2xz dz}{I} = 0$$

$$\frac{dI}{I} - \frac{2xz dy - 2xz dz}{xz(y+z)}$$

$$\frac{dI}{I} - \frac{2xz(dy + dz)}{xz(y+z)}$$

$$\log I - 2 \log(y+z) = \log c$$

$$I = c(y+z)^2$$

$$xy(y+z) = c(y+z)^2$$

$$\boxed{xy = c(y+z)} \quad \underline{\underline{\text{Ans}}}$$

Hom 3rd

③ $z^2 dx + (z^2 - 2yz) dy + (2y^2 - yz - xz) dz = 0$

Condition of integrability.

$$\boxed{Px + Qy + Rz = 0}$$

$$x = uz$$

$$y = vz$$

$$dz = z du + v dz$$

$$dy = z dv + v dz$$

$$xz^2 + (z^2 - 2yz)y + (2y^2 - yz - xz)z = 0$$

$$xz^2 + z^2y - 2y^2z + 2zy^2 - yz^2 - xz^2 = 0$$

$$Px + Qy + Rz = 0$$

6/9/25

Partial Differential equation

Charpit's method.

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} \quad (2)$$

$$f(x, y, z, p, q) = 0 \quad (1)$$

$$dz = p dx + q dy \quad (3)$$

Que ① function

$$2xz - px^2 - 2qxy + pq = 0 \quad (1)$$

$$\frac{dp}{2z - 2qy} = \frac{dq}{0} = \frac{dx}{x^2 - q} = \frac{dy}{2xy - p} = \frac{dz}{px^2 + 2xyq - 2pq} \quad (2)$$

$$dy = 0$$

$$q = a \text{ (const.)} \quad (3)$$

$$2xz - px^2 - 2qxy + pa$$

$$p(a - x^2) = 2axy - 2xz$$

$$p = \frac{2x(ay - z)}{a - x^2}$$

$$dz = \frac{2x(ay - z)}{a - x^2} dx + a dy$$

$$dz = p dx + q dy$$

$$dz - a dy = \frac{2x(ay - z)}{a - x^2} dx$$

$$\begin{cases} x^2 - a = t \\ 2x dx = dt \end{cases}$$

$$dz - a dy = \frac{2x(ay - z - ay)}{x^2 - a} dx$$

$$\frac{dz - a dy}{z - ay} = \frac{2x dx}{x^2 - a}$$

$$x^2 - a = t$$

$$\log(z - ay) = \frac{dt}{t}$$

$$\log(z - ay) = \log(x^2 - a) + \log c$$

$$\boxed{z - ay = (x^2 - a)b} \text{ Ans}$$

Que ② $(p^2 + q^2)y = qz$

$$\frac{dp}{p^2 + q^2} + q^2 y - qz = 0$$

$$f(x, y, z, p, q) = (p^2 + q^2)y - qz \quad \text{--- (1)}$$

$$\frac{dp}{-pq} = \frac{dq}{p^2} = \frac{dx}{p} = \frac{dy}{-q} = \frac{dz}{-q} =$$

$$\frac{dp}{-pq} = 0$$

$$\frac{dp}{-pq} = \frac{dq}{p^2}$$

$$\frac{dp}{p} = -q$$

$$\Rightarrow \frac{dp}{-q} = \frac{dq}{p}$$

$$\frac{p^2}{2} = -\frac{q^2}{2} + a'$$

$$\frac{p^2 + q^2}{2} = a' \Rightarrow p^2 + q^2 = a \quad \text{--- (3)}$$

$$dz = p dx + q dy$$

eq (3) in eqⁿ (1)

$$ay - qz = 0$$

$$\boxed{\frac{ay}{z} = q}$$

eqⁿ (3)

$$p^2 = a - y^2$$

$$p^2 = a - \frac{a^2 y^2}{z^2}$$

$$\boxed{p^2 = \frac{az^2 - a^2 y^2}{z^2}}$$

$$p = \frac{\sqrt{az^2 - a^2 y^2}}{z}$$

$$dz = p dx + q dy$$

$$dz = \frac{\sqrt{az^2 - a^2 y^2}}{z} dx + \frac{ay}{z} dy$$

$$z dz = \sqrt{az^2 - a^2 y^2} dx + ay dy$$

$$z dz - ay dy = \sqrt{az^2 - a^2 y^2} dx$$

$$\frac{z dz - ay dy}{\sqrt{az^2 - a^2 y^2}} = dx$$

$$az^2 - a^2 y^2 = t^2$$

$$2az dz - 2a^2 y dy$$

$$a(z dz - a y dy) = \frac{1}{2} dt$$

$$\frac{1}{a} \frac{dt}{t} = x + c \quad z dz - ay dy = \frac{1}{a} dt$$

$$\frac{1}{a} = x + c$$

$$\frac{\sqrt{az^2 - ay^2}}{a} = x + c \quad t = \sqrt{az^2 - ay^2}$$

Monge's method:

$$Rr + Ss + Tt = V$$

① Que $r - t \cos^2 x + p \tan x = 0$

imp most Que ② $r = a^2 t$

Ans ① $r - t \cos^2 x + p \tan x = 0$

$$Rr - Ss + Tt = V$$

$$R = 1 \quad S = 0 \quad T = -\cos^2 x$$

$$V = -p \tan x$$

Oscillatory eqⁿ, Ready text 1.
A.E.

$$R dy^2 + T dx^2 - S dx dy = 0$$

$$R p dy + T q dx - V dx dy = 0$$

Repdg Thanx ou very

$$1 dy^2 - \cos^2 x dx^2 = 0 \quad \text{--- (3)}$$

$$1 dp dy - \cos^2 x dq dx + p \tan x dndy = 0$$

2nd factor.

$$\text{--- (4)}$$

Eq (3) $[dy - \cos x dx] [dy + \cos x dx] = 0$ --- (5)

$$dy - \cos x dx = 0 \quad \text{--- (6)}$$

$$dp \cos x dx - \cos^2 x dq dx + p \tan x$$

$$y - \sin x = C_1 \quad \text{--- (7)}$$

$$dy + \cos x dx = 0 \quad \text{--- (8)}$$

$$dy = -\cos x dx$$

$$y + \sin x = C_2 \quad \text{--- (9)}$$

equation (6) put in eqn (4).

$$dp \cos x dx - \cos^2 x dq dx + p \tan x dx \cos x dx = 0$$

$$dp - \cos x dq + p \tan x dx \cos x dx = 0$$

$$dp + p \tan x dx - \cos x dq = 0$$

$$\sec x dp + p \sec x \tan x dx - dq = 0$$

$$p \sec x - q = C_2$$

$$C_2 = f(C_1)$$

$$p \sec x - q = f_1(y - \sin x) \quad \text{--- (10)}$$

2nd factor.

$$dy = -\cos x dx$$

$$dp(\cos x dx) - \cos^2 x dq dx + p \tan x dx (-\cos x dx)$$

$$- \cos dp \cos x dx - \cos^2 x dq dx - p \tan x dx \cos x dx = 0$$

$$p \sec x + q = C_4$$

$$C_4 = f_2(C_3)$$

$$C_4 = f_2(y + \sin x)$$

$$p \sec x + q = f_2(y + \sin x) \quad \text{--- (9)}$$

$$p \sec x - q = f_1(y - \sin x) \quad \text{--- (8)}$$

add

$$2p \sec x = f_1(y - \sin x) + f_2(y + \sin x)$$

$$p = \frac{f_1(y - \sin x) + f_2(y + \sin x)}{2 \sec x}$$

$$q = \frac{f_1(y - \sin x) - f_2(y + \sin x)}{2}$$

$$dz = \frac{f_1(y - \sin x) + f_2(y + \sin x)}{2 \sec x} dx + \frac{f_1(y - \sin x) - f_2(y + \sin x)}{2} dy$$

$$dz = f_1(y - \sin x) \left[\frac{dx}{2 \sec x} - \frac{dy}{2} \right] + f_2(y + \sin x) \left[\frac{dx}{2 \sec x} - \frac{dy}{2} \right]$$

$$+ f_2(y + \sin x) \left[\frac{dx}{2 \sec x} - \frac{dy}{2} \right]$$

integrate

$$z = F_1(y - \sin x) + F_2(y + \sin x) \quad \text{Ans.}$$

Ans $y+ax$ $y-ax$

imp
Que

$x^2 = a^2 t$

$z = f_1(y-ax) + f_2(y+ax)$